



GATE ME 90 Days Challenge

Day 5/90 · Practice Set with Detailed Solutions

SUBJECT

Strength of Materials

TODAY'S TOPIC

Stress, Strain & Mohr's Circle

COUNTDOWN

106 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on stress and strain, elastic constants, Poisson's ratio, Mohr's circle for plane stress and strain, and thermal stresses. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

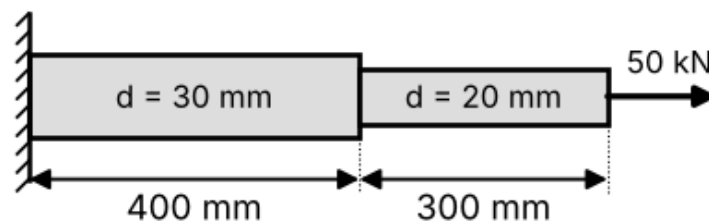
Q1 (MCQ)

In a tensile test on a 12 mm diameter steel specimen, a load of 20 kN produces an extension of 0.042 mm over a 50 mm gauge length (within the elastic limit). Young's modulus of the material is closest to

- (A) 210.5 GPa (B) 52.6 GPa
(C) 165.3 GPa (D) 200.0 GPa

Q2 (NAT)

The stepped steel bar shown is fixed at the left end and pulled by 50 kN at the free end. For $E = 200$ GPa, the total elongation (in mm) is _____



Q3 (MCQ)

For a homogeneous isotropic linearly elastic material, the theoretical range of Poisson's ratio is

- (A) 0 to 0.5 (B) -1 to 0.5
(C) 0 to 1 (D) -1 to 1

Q4 (NAT)

A material has $E = 200$ GPa and $G = 80$ GPa. Its bulk modulus K (in GPa) is _____

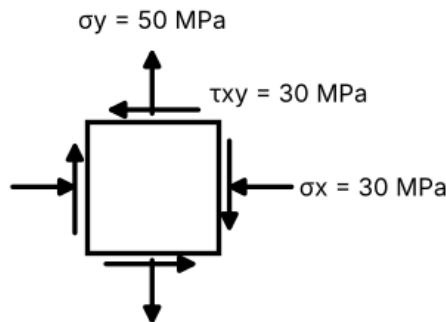
Q5 (MCQ)

A steel block $100 \text{ mm} \times 50 \text{ mm} \times 20 \text{ mm}$ is subjected to $\sigma_x = 100 \text{ MPa}$ (tension) along the 100 mm side and $\sigma_y = 50 \text{ MPa}$ (compression) along the 50 mm side; $\sigma_z = 0$. For $E = 200 \text{ GPa}$ and $\nu = 0.3$, the change in volume of the block is

- (A) increase of 25 mm^3 (B) decrease of 10 mm^3
 (C) increase of 4 mm^3 (D) increase of 10 mm^3

Q6 (NAT)

For the plane stress element shown ($\sigma_x = 30 \text{ MPa}$ compressive, $\sigma_y = 50 \text{ MPa}$ tensile, $\tau_{xy} = 30 \text{ MPa}$ in the directions shown), the maximum in-plane shear stress (in MPa) is _____

**Q7** (MCQ)

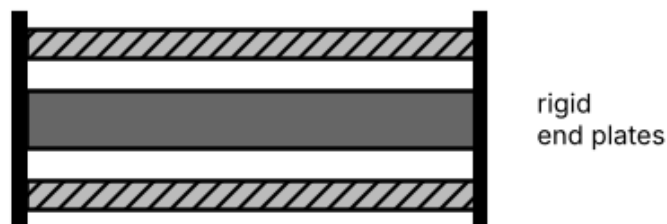
At a point, the principal stresses are 100 MPa and 40 MPa (both tensile, plane stress). On the plane of maximum in-plane shear stress, the normal stress and the shear stress are

- (A) 0 and 30 MPa (B) 0 and 70 MPa
 (C) 70 MPa and 30 MPa (D) 30 MPa and 50 MPa

Q8 (NAT)

A steel rod is placed inside a copper tube of the same length and the two are joined rigidly at the ends, as shown. The assembly is heated by 80°C . The stress in the steel rod (in MPa) is _____

Copper tube: $A = 1000 \text{ mm}^2$, $E = 100 \text{ GPa}$, $\alpha = 17 \times 10^{-6} / ^\circ\text{C}$



Steel rod: $A = 500 \text{ mm}^2$, $E = 200 \text{ GPa}$, $\alpha = 12 \times 10^{-6} / ^\circ\text{C}$

Q9 (MCQ)

A thin plate is in pure shear with $\tau = 60 \text{ MPa}$. For $E = 200 \text{ GPa}$ and $\nu = 0.25$, the major principal strain is

- (A) 300×10^{-6} (B) 225×10^{-6}
 (C) 375×10^{-6} (D) 750×10^{-6}

Q10 (NAT)

At a point in plane strain, $\epsilon_x = 800 \times 10^{-6}$, $\epsilon_y = 200 \times 10^{-6}$ and $\gamma_{xy} = -800 \times 10^{-6}$. The major principal strain (in units of 10^{-6}) is _____

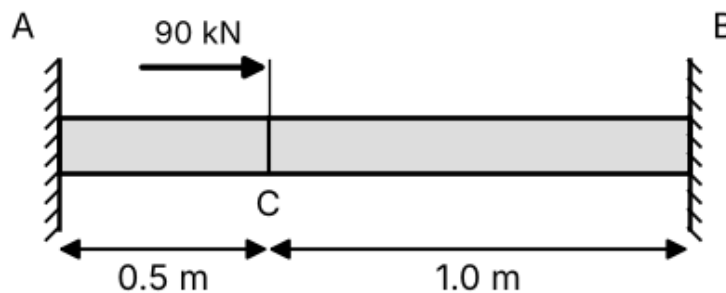
Q11 (MCQ)

A conical bar and a prismatic bar of the same length and material hang vertically from their tops (the cone from its base). The ratio of the elongation of the conical bar to that of the prismatic bar under their own weights is

- (A) $1/2$ (B) $1/3$
(C) $1/6$ (D) $2/3$

Q12 (NAT)

The bar AB shown is fixed at both ends and carries an axial load of 90 kN at C. The reaction at the end A (in kN) is _____

**Q13** (MCQ)

A block is loaded by $\sigma_x = 100$ MPa (tension) while it is fully restrained against strain in the y-direction ($\epsilon_y = 0$); the z faces are free ($\sigma_z = 0$). For $E = 200$ GPa and $\nu = 0.3$, the strain ϵ_x is

- (A) 500×10^{-6} (B) 455×10^{-6}
(C) 545×10^{-6} (D) 350×10^{-6}

Q14 (NAT)

At a point, $\sigma_x = 100$ MPa and $\sigma_y = 20$ MPa (both tensile). If the major principal stress must not exceed 120 MPa, the largest permissible shear stress τ_{xy} (in MPa) is _____

Q15 (MCQ)

At a point in a body under plane stress, $\sigma_x = \sigma_y = 60$ MPa and $\tau_{xy} = 0$. Which statement is correct?

- (A) The maximum in-plane shear stress is 60 MPa (B) The principal planes are at 45° to x
(C) The maximum in-plane shear stress is 30 MPa (D) Mohr's circle is a point; every plane is a principal plane

Answer Key

Q	Type	Answer
Q1	MCQ	(A) 210.5 GPa
Q2	NAT	0.380 (accept 0.375 to 0.385)
Q3	MCQ	(B) -1 to 0.5
Q4	NAT	133.3 (accept 132.5 to 134.2)
Q5	MCQ	(D) increase of 10 mm ³
Q6	NAT	50
Q7	MCQ	(C) 70 MPa and 30 MPa
Q8	NAT	40 (tensile)
Q9	MCQ	(C) 375×10^{-6}
Q10	NAT	1000
Q11	MCQ	(B) 1/3
Q12	NAT	60
Q13	MCQ	(B) 455×10^{-6}
Q14	NAT	44.72 (accept 44.4 to 45.0)
Q15	MCQ	(D) Mohr's circle is a point; every plane is a principal plane

Detailed Solutions

Q1 Answer: (A) 210.5 GPa

Concept: In the elastic range, $E = \text{stress/strain} = PL/(A\delta)$.

$$A = (\pi/4)(12^2) = 113.1 \text{ mm}^2$$

$$E = (20000 \times 50)/(113.1 \times 0.042) = 2.105 \times 10^5 \text{ N/mm}^2 = \mathbf{210.5 \text{ GPa}}$$

Exam tip: Use the gauge length, not the total specimen length, and $A = \pi d^2/4$ (using πd^2 gives the 52.6 GPa trap).

Q2 Answer: 0.380 (accept 0.375 to 0.385)

Concept: Each segment carries the full 50 kN; add the elongations of the segments in series: $\delta = (P/E) \sum L/A$.

$$A_1 = (\pi/4)(30^2) = 706.9 \text{ mm}^2, A_2 = (\pi/4)(20^2) = 314.2 \text{ mm}^2$$

$$\delta_1 = 50000 \times 400/(706.9 \times 200000) = 0.1415 \text{ mm}$$

$$\delta_2 = 50000 \times 300/(314.2 \times 200000) = 0.2387 \text{ mm}$$

$$\delta = \mathbf{0.380 \text{ mm}}$$

Exam tip: The thinner segment stretches more even though it is shorter: area enters as d^2 .

Q3 Answer: (B) -1 to 0.5

Concept: Both G and K must be positive for a stable material.

$G = E/[2(1 + \nu)] > 0$ needs $\nu > -1$; $K = E/[3(1 - 2\nu)] > 0$ needs $\nu < 0.5$.

So $-1 < \nu < 0.5$. Most engineering metals lie between 0.25 and 0.35; cork is close to 0 and rubber close to 0.5.

Exam tip: "0 to 0.5" is the practical range; GATE asks the theoretical one.

Q4 Answer: 133.3 (accept 132.5 to 134.2)

Concept: $E = 2G(1 + \nu) = 3K(1 - 2\nu)$.

$\nu = E/(2G) - 1 = 200/160 - 1 = 0.25$

$K = E/[3(1 - 2\nu)] = 200/(3 \times 0.5) = \mathbf{133.3 \text{ GPa}}$

Exam tip: Direct form: $K = GE/[3(3G - E)] = 80 \times 200/(3 \times 40) = 133.3 \text{ GPa}$.

Q5 Answer: (D) increase of 10 mm³

Concept: Volumetric strain $\epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z = (1 - 2\nu)(\sigma_x + \sigma_y + \sigma_z)/E$.

Sum of stresses = $100 - 50 + 0 = 50 \text{ MPa}$

$\epsilon_v = 0.4 \times 50/200000 = 1.0 \times 10^{-4}$

$V = 100 \times 50 \times 20 = 10^5 \text{ mm}^3$, $\Delta V = \mathbf{+10 \text{ mm}^3}$ (increase, since the sum is tensile)

Exam tip: Forgetting $(1 - 2\nu)$ gives 25 mm^3 .

Q6 Answer: 50

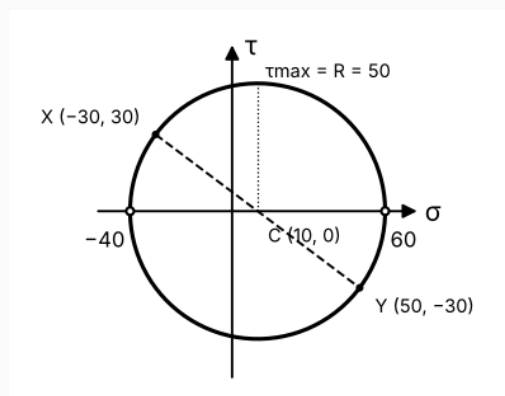
Concept: Mohr's circle: centre $C = (\sigma_x + \sigma_y)/2$, radius $R = \sqrt{[(\sigma_x - \sigma_y)/2]^2 + \tau_{xy}^2}$; $\tau_{\max, \text{in-plane}} = R$.

$C = (-30 + 50)/2 = 10 \text{ MPa}$

$R = \sqrt{40^2 + 30^2} = \mathbf{50 \text{ MPa}}$; principal stresses $\sigma_1 = 60 \text{ MPa}$, $\sigma_2 = -40 \text{ MPa}$.

Points: $X(-30, 30)$ and $Y(50, -30)$, with clockwise shear plotted upward; XY is a diameter.

Exam tip: Here σ_1 and σ_2 have opposite signs, so the absolute maximum shear equals the in-plane value R .



Q7 Answer: (C) 70 MPa and 30 MPa

Concept: The planes of maximum in-plane shear are at 45° to the principal planes; on them σ_n = average stress.

$$\tau_{\max} = (100 - 40)/2 = 30 \text{ MPa}$$

$$\sigma_n = (100 + 40)/2 = 70 \text{ MPa} \rightarrow \mathbf{70 \text{ MPa and 30 MPa}}$$

Exam tip: Maximum shear planes are NOT free of normal stress (unlike principal planes, which are free of shear).

Q8 Answer: 40 (tensile)

Concept: Copper wants to expand more than steel. The joint forces a common extension, so steel goes into tension and copper into compression. Force balance + compatibility.

$$\text{Force balance: } \sigma_s A_s = \sigma_c A_c \rightarrow \sigma_c = \sigma_s (500/1000) = 0.5 \sigma_s$$

$$\text{Compatibility: } \sigma_s/E_s + \sigma_c/E_c = (\alpha_c - \alpha_s)\Delta T = 5 \times 10^{-6} \times 80 = 4 \times 10^{-4}$$

$$\sigma_s(1/200000 + 0.5/100000) = 4 \times 10^{-4} \rightarrow \sigma_s \times 10^{-5} = 4 \times 10^{-4}$$

$$\sigma_s = \mathbf{40 \text{ MPa (tension)}}; \sigma_c = 20 \text{ MPa (compression)}$$

Exam tip: The material with the larger α always ends up in compression.

Q9 Answer: (C) 375×10^{-6}

Concept: Pure shear τ is equivalent to $\sigma_1 = +\tau$ and $\sigma_2 = -\tau$ at 45° .

$$\epsilon_1 = (\sigma_1 - \nu\sigma_2)/E = \tau(1 + \nu)/E = 60 \times 1.25/200000$$

$$\epsilon_1 = \mathbf{375 \times 10^{-6}}$$

$$\text{Check: } \gamma = \tau/G \text{ with } G = 80 \text{ GPa gives } \gamma = 750 \times 10^{-6}, \text{ and } \epsilon_1 = \gamma/2 = 375 \times 10^{-6}. \checkmark$$

Exam tip: 300×10^{-6} ($= \tau/E$) forgets the Poisson effect of the compressive principal stress.

Q10 Answer: 1000

Concept: The strain Mohr's circle uses ϵ on the horizontal axis and $\gamma/2$ on the vertical axis.

$$\text{Centre} = (800 + 200)/2 = 500$$

$$\text{Radius} = \sqrt{[(800 - 200)/2]^2 + (-800/2)^2} = \sqrt{300^2 + 400^2} = 500$$

$$\epsilon_1 = 500 + 500 = \mathbf{1000 \times 10^{-6}}, \epsilon_2 = 0$$

Exam tip: Always halve γ before using it; the sign of γ only changes the angle, not the principal values.

Q11 Answer: (B) 1/3

Concept: Elongation under self-weight: prismatic bar $\delta = \rho g L^2/(2E)$; conical bar $\delta = \rho g L^2/(6E)$.

$$\text{Ratio} = (1/6)/(1/2) = \mathbf{1/3}$$

The cone is lighter near the free tip, so each section carries less weight below it.

Exam tip: Both results are independent of the cross-sectional area.

Q12 Answer: 60

Concept: Statically indeterminate: equilibrium + compatibility (total change of length $AB = 0$).

Equilibrium: $R_A + R_B = 90 \text{ kN}$

Part AC (0.5 m) is in tension by R_A ; part CB (1.0 m) is in compression by R_B . Same A and E:

$$R_A(0.5) = R_B(1.0) \rightarrow R_A = 2R_B$$

$$R_B = 30 \text{ kN}, R_A = \mathbf{60 \text{ kN}}$$

Exam tip: The nearer support takes the larger share: $R_A = P \times CB/AB$.

Q13 Answer: (B) 455×10^{-6}

Concept: Generalised Hooke's law with a restraint: first find the induced stress, then the strain.

$$\epsilon_y = (\sigma_y - \nu\sigma_x)/E = 0 \rightarrow \sigma_y = 0.3 \times 100 = 30 \text{ MPa (tensile)}$$

$$\epsilon_x = (\sigma_x - \nu\sigma_y)/E = (100 - 9)/200000 = \mathbf{455 \times 10^{-6}}$$

Exam tip: 500×10^{-6} ($= \sigma_x/E$) ignores the restraint. A lateral restraint makes the material look stiffer.

Q14 Answer: 44.72 (accept 44.4 to 45.0)

Concept: $\sigma_1 = C + R$ with $C = (\sigma_x + \sigma_y)/2$ and $R = \sqrt{[(\sigma_x - \sigma_y)/2]^2 + \tau^2}$.

$C = 60 \text{ MPa}$, so $R \leq 120 - 60 = 60 \text{ MPa}$

$$\sqrt{(40^2 + \tau^2)} \leq 60 \rightarrow \tau^2 \leq 3600 - 1600 = 2000$$

$$\tau_{\max} = \mathbf{44.72 \text{ MPa}}$$

Exam tip: Work backwards from the allowable on Mohr's circle: fix the centre, then the radius.

Q15 Answer: (D) Mohr's circle is a point; every plane is a principal plane

Concept: Radius $R = \sqrt{[(\sigma_x - \sigma_y)/2]^2 + \tau_{xy}^2} = 0$.

So Mohr's circle shrinks to the single point (60, 0): every plane through the point has $\sigma_n = 60 \text{ MPa}$ and $\tau = 0$.

Hence **every plane is a principal plane** and the in-plane shear stress is zero everywhere. (The out-of-plane maximum shear is still 30 MPa, since $\sigma_z = 0$.)

Exam tip: Equal biaxial tension, like the wall of a thin spherical shell, is the classic example.

What next?

Day 5 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124



Website test series



Dig Career Thrust app



Telegram: daily PDFs

Next, Day 6/90: Week 1 Revision, a weekly test on all five topics of this week. Comment **GATE90** on our reel to get all links.

Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.